A Historic El Niño Is Coming That Could Cost the World Trillions

Podcast: Odd lots

AI Executive Summary

1. Core Thesis: The episode argues that El Niño is not merely a cyclical weather event causing temporary economic price shocks, but a systemic force capable of permanently depressing global economic growth trajectories due to its synchronous impact across disparate regions and the compounding effect of global warming. The current upcoming El Niño event is forecast to be record-breaking and will occur during a period of heightened economic fragility, potentially causing trillions of dollars in losses within five years (estimated conservatively at $10 trillion up to $14 trillion) through mechanisms that disrupt supply chains, reduce productivity, and create systemic uncertainty that markets have yet to fully adapt to.

2. Key Arguments & Data:

3. **Physical Mechanism:** El Niño is a mode of variability within the El Niño Southern Oscillation (ENSO), defined by a weakening of easterly trade winds in the tropical Pacific causing surface warm water (the West Pacific warm pool) to slosh eastward, exposing more global atmospheric energy to heat from the ocean.
4. **Forecast Accuracy & Magnitude:** While forecasts vary, models are converging on a potential record-breaking El Niño since 1870s (compared to earlier 20th-century benchmarks). The Relative Oceanic Nino Index is projected to exceed 3.3 thresholds. Every forecast places the current event as warmer than the 2016 El Niño.
5. **Economic Impact - Growth vs. Levels:** Contrary to historical understanding where El Niño is viewed as a level effect (shocks to prices that recover), a new study published in *Science* (2023) by Mankin's lab indicates a systematic depression of economic growth. This loss accrues ad infinitum because the simultaneous nature of hazards across regions prevents economies from normalizing trajectories back to pre-shock levels.
6. **Quantifying Losses:** The 1997-98 El Niño generated approximately $36 billion in immediate economic losses (level shock). However, over a five-year period following that event, cumulative global losses were

estimated at about $5.7 trillion due to permanent growth trajectory shifts.

7. **Global Warming & Out-of-Sample Variance:** Current El Niño forecasts are "out of sample" because they occur superimposed on background global warming trends. The ocean acts as a heat sink, storing excess greenhouse gases; this increases the baseline energy available during El Niño events.
8. **Projected Record Temperatures:** 2026 has a non-trivial chance of being the warmest year on record not just due to anthropogenic warming but due to this specific El Niño. 2027 is projected to be virtually guaranteed as a record temperature year despite the event's lifecycle.
9. **Resilience Gaps:** Evidence suggests humans are not optimally adapted to El Niño variability despite centuries of exposure. Current market inefficiencies and lack of coordinated global strategy contribute to unmitigated economic tolls compared to potential adaptation strategies seen locally (e.g., Peruvian potato farmers forecasting based on Pleiades visibility).

10. **Endogeneity:** The relationship between climate variability and the economy is endogenous; climate affects growth, but greenhouse gas emissions induced by that growth alter the climate trajectory, creating a feedback loop.

11. Actionable Takeaways:

12. **Monitor El Niño Cycles:** Investors and policymakers should treat El Niño as a natural experiment or "treatment" distinct from general weather volatility to forecast economic productivity deviations rather than just price level shocks.
13. **Prepare for Systemic Risk:** Recognize that the current El Niño is an out-of-sample event superimposed on global warming; strategies focused solely on historical averages may be insufficient to mitigate risks in 2026-2027.
14. **Diversify Against Hazards:** Utilize risk diversification and early warning systems (e.g., insurance markets, emergency response teams)

to protect capital-intensive assets against cascading hazards like droughts, floods, and road washouts typical of El Niño regions.

15. **Account for Permanent Losses:** Financial models should incorporate the possibility that economic growth may be permanently depressed following major El Niño events rather than assuming standard recovery timelines.
16. **Investigate Adaptation Strategies:** Explore and implement resilience strategies that go beyond local infrastructure (like reservoirs) to include global market efficiency, improved forecasting, and coordination mechanisms to handle simultaneous regional hazards.

Full Transcript

A new chapter in global growth is being written and much of it is happening in Africa. Africans need to invest. There are deals to be done and business to be won. I'm Jennifer Zabasajap. Every week on the Next Africa podcast, we track capital flows and political shifts shaping the continent's future. The digitalization of Africa is going to power its growth. Reading the world of something like HIV is possible. Population growth is so enormous in Africa. Listen to Next Africa on Apple's Spotify or wherever you get your podcast. Bloomberg Audio Studios. Podcasts, radio, news. Hello and welcome to another episode of The Odd Thoughts Podcast. I'm Tracy Allaway. And I'm Joe Wiesenthal. Joe, I don't know if I've ever told you about this, but I'm a little bit of a weather nerd. Okay, I did maybe a little, but I knew. I just get really excited by weather. I love thunderstorms. I love a good thunderstorm. So great. If I'm inside. Whoa, this is the thing, I love thunderstorms except when I'm on an airplane or something like that.

I do not like economically damaging weather. I feel like I need to throw that caveat in there. You know what I am. I'm like a weather truther. So, for example, I don't believe that there's any accurate way, for example, to measure wind chill. It's like, oh, it's like 25 degrees out and it feels like 15. It's like, how do you know what 15 feels like? You know what I'm saying? Like, I don't believe in some of these numbers. Also, humidity, what, there's water in the air? Give me a break. There's no water in there. No, that one I believe in. Tell me you've never been to Abu Dhabi in the summer without telling me you've never been to Abu Dhabi in the summer. No, this last week in New York. You were just in Hong Kong, you know what humidity is.

No, that one was a joke. I am a windshield skeptic, even though I don't think that's the topic of our conversation today. It is not. The topic of our conversation is actually economically damaging weather. Yes. And specifically the upcoming El Nino, which, by the way, it's not that... That often that you see a lot of sell side notes start to mention whether phenomenon but just in the past you know a few days I've had a Thailand economics note from city talking about interest rates and quoting the

potential impact of El Nino potentially inflationary I've had a Bank of England outlook from Morgan Stanley that mentioned El Nino and also a Bank of America FX outlook talking about the impact on EM currencies. So this is something that people are expecting to have an actual impact both on the global economy and the investment landscape. You know, I will just say, even though I'm not a weather nerd the way you are, I sort of have this thought that if you look at like the really grand sweep of history, like thousands and thousands of years, maybe even hundreds of thousands of years, I sort of think there are two things that matter in human history. One of them, as you know, I'm interested in is like, basically technological evolution, media technology, and how it reshapes how we think. And then the other one is sort of climate change, weather and storms. And I think that when we think about history, many things are contingent either on like brief or long periods of weather change and it can reshape society in permanent, long-term, unexpected ways at the time. And have we ever even done any El Nino episode in like 10 years or 11 years of doing this? I don't think we have. But it sounds like this is one that we should cover if it's showing up on all these notes. If we're going to cover one, this should probably be the one because at the moment, well, you have different El Nino forecasts. Yeah. And, you know, there's a range of probability.

involved here, but a lot of the forecasting models are currently forecasting that this is going to be a potentially record-breaking El Nino, something we haven't seen in decades. Do you think there are any people who are like, oh, it's interesting you bring up the El Nino? You know, I've always thought La Ninos are the more underrated one. Do you think that's a type of guy? No, I hope not. But are you going to become like an El Nino truther and you're going to be like, how do you know it's El Nino unless you're standing in the Pacific Ocean measuring the water yourself? I'm going to become a guy who's like, you know, you really should be paying attention to the upcoming La Nina. Oh, good. That'll be a fun transformation for you. OK, well, we do, in fact, have the perfect guest. We're going to be speaking to someone who not only follows the weather but also tries to measure its economic impact. We have Justin.

Mankin, he is Professor of Geography at Dartmouth and Director of the Climate Modeling and Impacts Group. So Justin, thank you so much for

coming on OddLods. Yeah, happy to be here. Thanks for having me. I feel like maybe the first thing we should do is, can you convince Joe that El Nino is a real thing? No, apparently he needs like a first person experience of it. What is El Nino? Yeah, what is it? Yeah, light windchill. factor. It is, it is in fact real and measurable. And they're both kind of actually aerodynamically related, right? So windshield, the feeling you get on your skin is going to the fact that the wind is wicking away latent heat from your skin, which is the heat associated with a phase change of water from a liquid to a vapor. And so that's why when you get out of a shower, you feel cooler.

than before you get into it, right? So that evaporation off your skin induced by aerodynamic resistance, i.e. the wind, that is the thing that's cooling you down, which is why we can measure it and why we can tell you that cold exposure is a bad thing. El Nino phenomenon is an ocean atmosphere. It's kind of part of what's called the El Nino Southern Oscillation, which is essentially a relationship between the tropical Pacific and the overlying tropical atmosphere. And it has a tendency. to be in what we call a neutral state, which is kind of average conditions where you have these easterly trade winds. So these are winds that are blowing from east to west across the tropics along the equator. And what they're doing is they're dragging surface water, surface water that's been worn because it's in the tropics. The sun is just shining there all year round. And so that water heats up and then the winds kind of drag that surface water and create literally a pile.

such that the sea altitude is higher in the West Pacific than in the East Pacific. And that pile of warm water is something we call the West Pacific warm pool. And so it just sits there parked by Indonesia and Australia. And then every once in a while those trade winds weaken and break down. And that pile of warm water that actually penetrates quite deep into the western tropical Pacific. That pile of warm water just spreads itself out over the vast expanse that is the tropical Pacific. And so that's actually how we measure El Nino is we're measuring it as a function of sea surface temperature anomaly. So how much warmer than average are ocean temperatures at the surface?

you know, at this time in this location. And you know, there are different ways to kind of calculate what the regional average is and we can get into that technical nitty gritty. But essentially all this water sloshes back eastward and in doing so reconfigures all of weather, which is just energy redistribution from the sun on our fluid planet, right? It redistributes all of that energy differently than would have been the case if all that water were piled up. in the western Pacific. I guess like a really simple analogy a listener could maybe grab onto is they've walked into a cold room where there's a little space heater plugged into the corner, right? And that little space heater, yeah, is maybe generating some heat that's warming up the room, but it's really only warming up that corner of the room. That is akin to neutral.

or what are called La Nina-like conditions, which is an amplification of the neutral pattern, meaning that those winds are stronger than normal and the eastern tropical Pacific is cooler than normal. When you slosh all of that water back across the entire spans of the tropical Pacific, it's akin to walking into a room that has wall-to-wall radiators instead. the thermodynamics of that room, that's a totally different experience for you than would have been the case if it was just that space heater in the corner. And so when an El Nino occurs, what it is doing, it is essentially the tropics are the heat engine of global weather and climate, and it's a complete reconfiguration such that so much more of our global atmosphere is exposed to that heat energy from the ocean that would have otherwise been buried in the western tropical Pacific.

Does that make sense? That was an amazing answer. I understand something that I went 45 years without having any idea of, and now I get it. And as someone who used to live in a drafty New England home, I have a lot of experience with space heaters that only heat up that corner of the room. What are the conditions or what are the drivers such that these El Niños are semi-regular? Why do they seem to come? every few years? Why are they not a permanent pattern? Why is one year an El Nino year and another year isn't? Yeah, there's a lot of work on kind of the drivers of the breakdown of the winds in the eastern tropical Pacific. I think a simple answer is that we live on a sphere that's rotating and has stratified and unstratified fluids in the form of the ocean and the atmosphere. And that

essentially means that energy is propagated as wave phenomena, right, with peaks and troughs. And El Nino is akin to a wave phenomena, right? It is kind of a pattern, what we call a mode of variability. George Philander, who is this Princeton geophysicist and major scholar of El Nino called El Nino the trunk of the variability tree when it comes to climate variation in terms of kind of regulating these pulses of energy redistribution around our planet and so you know a phenomena like El Nino being modal meaning it has a canonical pattern temporally and spatially and then induces physical responses that we can characterize and use as a means of prediction right like we know what we should expect for Peru. We know what we should expect for Australia. And we're able to do that because we've diagnosed this mode of variability and given it a name and characterized its length scales and characterized what we call its teleconnections, meaning the kind of physical bridge between El Nino as a phenomenon and what it implies for the Indian monsoon.

or for the southwestern United States temperature and precipitation regime for that season. And we've given these things names and they've come instead of being purely a source of kind of stochastic uncertainty, just kind of random variation in weather and climate, they become a means of predictability. And that's what's really powerful about investigating these type of phenomena and giving them names, right? We have El Nino or El Nino Southern Oscillation, which is the kind of teleconnection with the overlying atmosphere. We have the North Atlantic Oscillation. We have the Indian Ocean Dipole. We have the Matt and Juliet Oscillation. So we've identified these patterns and their kind of manifestation. And in terms of like their underlying drivers, which is I think the question you're asking, I think it's just a function mostly of living on a complex planet with geophysical fluid dynamics and such that these wave phenomena just kind of emerge on these characteristic time scales. The precise mechanism of action that kind of leads the breakdown.

of these easterly trade winds, which like triggers what's called the birkeness feedback, which is this positive feedback loop. So an amplifying feedback that tends to reinforce itself can come from myriad different places from changes in ice cover to a weakening of the equator to pole temperature gradient. All kinds of different things have some culpability and potentially

causing a breakdown of these winds that allow the sea surface temperatures to slosh eastward again and reconfigure all severe weather on our planet. Talk to us about what the forecast actually is for this particular El Nino, because I read these things like, apparently, the Australian Bureau of Meteorology, which would be interested in this because Australia is particularly affected by El Nino, apparently, the relative oceanic Nino index.

Or Ronnie for short apparently has never reached the 3.3 thresholds that they currently project before so that seems extreme What are what exactly are we expecting here? So there's a bunch of different ways to forecast El Mino all of them kind of stem from some of the original innovative work in the late 70s and early 80s by Mark Kane and Steve Zeviak who built the first kind of physical model of El Nino to explain it as a phenomenon and then actually make predictions of its occurrence which is pretty remarkable but all these are kind of based on some of them are statistical in nature so kind of taking temperatures at time t and making a prediction given kind of autocorrelation in that time series, what it should look like at time t plus one. There are others that are physical models that are actually simulating the relationship between oceanic temperatures, winds, the overlying atmosphere, what it does to the kind of pressure and circulation that actually gets air moving in the first place. And so those ones are called process based models. And those are distinct from these statistical models, but all of them, regardless of how being modeled are kind of converging on a record El Nino at least since 1950 but arguably since much earlier right the 1877-1878 El Nino which was associated with the great famines and Mike Davis has this amazing book from 2002 about that El Nino event that One I think is, you know, kind of indistinguishable and strength from El Nino as we've already had, such as the ones in 1982 and 83, the 97 and 98 El Nino, which listeners may remember as being associated with Chris Farley's SNL skit of I am the El Nino, right? And, you know, the 2015 and 2016 El Nino, which I believe is the one on record depending on your indices. So we've had really strong.

El Ninos but this one is looking like it's much bigger potentially. Every forecast right now I believe puts it forthcoming the unfolding El Nino because we are in a in an advisory and we have been since June according

to the National Oceanic and Atmospheric Administration's Climate Prediction Center. Essentially you're looking at every forecast being having a warmer El Nino than the 2016 El Nino. Our hometown is not a test tube. 90 miles northeast of Nashville, a battle for the future of America plays out in one small town. Developers with right-wing ties have purchased hundreds of acres of land. We need cities on a shining hill. This is Our Town, a podcast about what happens when a small town becomes the site of a social experiment and fights back. I guess you didn't move in on a bunch of dumb hillbillies now, did you? Listen to Our Town on the iHeart radio app.

Apple podcasts or wherever you get your podcasts. How did we actually discover El Nino as a weather phenomenon? Because it happens with some regularity, but it's not like it happens every three to four years, like right on time at the same intensity as the previous round. So how did people identify this as an actual measurable phenomenon? You know in terms of kind of like the modern instrumental and climate science era I don't actually know the answer to that, but I do know something that's actually I'd argue even more interesting. So the person who first modeled El Niño from a physics perspective, Mark Kane, some amazing geophysicist at La Montdority Earth Observatory at Columbia University, and his colleagues, John Cheng, and other climate scientists there at the time, and then an anthropologist by the name of Ben Orloff, who is an anthropologist at Columbia, they wrote this incredible paper, I believe in nature, back in the...

90s or early aughts where they essentially showed that there was really good archaeological evidence that Peruvian potato farmers as long as 500 years ago were forecasting El Nino and they were doing it based on the visibility of the Pleiades which is a star system you can see from the Andes and essentially as El Nino occurs, what it's doing is it's reconfiguring the distribution of high versus low clouds in the Eastern tropical Pacific. It's essentially taking convection that was happening near Australasia and moving it eastward. And in doing so, you're creating all these kind of high-series cloud shields over the Andes, and that reduces the visibility. of this star pattern. And they would alter their planting dates based on that. So there's just kind of this remarkable cultural knowledge of being able to grab onto something and be able to make a forecast, an informed forecast. And

so I'd argue there's a lot of cultural knowledge about these innate climate phenomena around the world that we have yet to kind of appreciate and diagnose in the highly industrialized world where we're so.

relying on instrumentation. Yeah, folk meteorology or folk weather is like extraordinary and like how much they're able to anticipate. I was sort of, I was, you know, I imagined that there were probably various civilizations that interpreted on Nino's like the king has lost the mandate of heaven. And now, you know, there's a drought. Now we have to have a new political order. Also, this Mike Davis book that you mentioned looks fascinating and sort of speaks to the point of like, late Victorian holocausts, El Nino famines, and the making of the third world talking about subsequent famines in India, Northern China, and Northeastern Brazil. Definitely going to have to read that book. When we talk about a record El Nino, what is the measure? Is it the heat variation? Is it the expected change in rainfall or drought? Like what are the sort of the gauges that we use to say this is a record El Nino or not or measuring that? What are what are the parameters of an El Nino?

My understanding is it's all benchmarked to the magnitude of the sea surface temperature anomaly. An anomaly is a departure from expectation or departure from normal. And so what constitutes normal becomes really important there for setting the magnitude of how anomalous something is. And that's why, and Tracy was talking about, the relative Nino index, the Rony. right that's kind of shifting kind of a rolling baseline because we have secular warming in the tropical pacific owing to global warming that's occurring right as you heat up the planet the largest reservoir of heat storage just happens to be the 70 volume of water that constitutes our planet surface and it takes a lot of energy to heat up a volume of water. It's much more energy intensive to heat up a volume of water than it is an equivalent volume of air. So that specific heat is just much higher. So it's just a great energy reservoir and the ocean has been doing humanity an incredible solid by absorbing all this excess heat, which is the problem of global warming, right? That we're not allowing our atmosphere to shed heat as efficiently as it otherwise would have been if greenhouse gases We're lower in our atmosphere and in doing so.

we're having the heat up to reattain equilibrium such that the energy coming in from the Sun is balanced by Earthlight leaving out to space. That is not in balance at the moment and a lot of that excess heat is going into the ocean and that's how planetary temperatures increase such that in can equal out again. That's essentially a very simple global warming 101 but the way we're measuring El Nino as a phenomenon is by taking a regional average to choosing an area in the tropical Pacific that is canonically associated with the manifestation of El Nino. And you can choose different regions and then benchmarking those average temperatures in this month to expectation what have average has been.

Sure. Over time. And so you can choose different regions. So there's a Nino 3 region, which has kind of shifted towards the central Pacific. There's a Nino 4 region, which has shifted a bit more eastward. There's a Nino 3-4 region, which is kind of the typical region that's now used, which characterizes El Nino over something in between those two regions. Then there's the E-index, which is taking a dimension reduction technique on a spatial field and collapsing it to say these are kind of the dominant directions of variability in that field. And that E index would kind of represent something differently in the sense that El Nino, while we think of it as a singular phenomenon, it has a geography associated with it. And that geography means that it has flavors associated with it. And so these flavors mean that different parts of the tropical Pacific warm differently and look anomalously different. And so not every El Nino is the same. And so it's consequences in terms of the hazard it brings for the United States or for hurricane occurrence in the Atlantic basin or whatever. It's not the same for every El Nino event. The spatial pattern of the sea surface temperature anomalies also seems to matter. And so there's been a lot of work on kind of diagnosing these different flavors of El Nino, and the e-index is intended to try to capture some of that. So there are lots of different ways of measuring it, but it's always a sea surface temperature anomaly in degrees Celsius benchmark to some expectation or average of normal. So one of the things that Joe and I have learned in doing various agricultural episodes over the years is that when you have a supply shock, the timing matters.

Right? So you generally don't want to declare war in the spring planting season when all your soldiers might be off planting stuff instead. Does the

timing of this particular El Nino matter? I keep reading that the peak of this El Nino is expected to be in the fall in the Northern Hemisphere. Is that particularly unfortunate timing? You know, there's been a lot of work that looks, you know, the planting and harvest of crops tends to be something that is highly calendar dependent and there's a bit of flexibility that I think farmers have to be responsive to their expectations for weather variability. And certainly ag forecasts have become important and agriculture at large has become really dependent on on El Nino forecast. The thing about El Nino is that it does typically peak in December, January and February. So this is kind of the boreal meaning the northern hemisphere winter. And that is implications for the austral summer, right, for the southern hemisphere summer, which is in the opposite season. And also, there's also kind of a time dependency associated with the impact. So let me just concretize this with an example. We've been in an El Nino advisory since June. That's also coincident with the typical onset of the South Asian monsoon, which is a determinant of these for famine for billions of people, right? You can link monsoon onset to, you know, farmer well-being and farmer suicides. You can link the timing of monsoon onset to diesel prices within India and the political fates of people around diesel subsidies because if the monsoon onset is delayed, you have a bunch of farmers suddenly needing to irrigate crops where they otherwise wouldn't have had to and that that irrigation is dependent on the generators that rely on diesel fuel. And so you can see price spikes around those types of things and the political fortunes of like municipal and kind of local politicians kind of surviving or not surviving elections based on those kinds of things. So there's like lots of cascading ways in which in El Nino, even at this early stage in its life cycle.

which tends to kind of start after what we call the spring prediction barrier, which is kind of this period in time where it's difficult to make a prediction about whether or not an El Nino is going to occur to next February. So we're looking at kind of a wind down of this event in February, March of 2027. What's interesting about it is that it's going to lock in all sorts of phenomena. 2026 has a non-trivial chance of being the warmest year on record. not simply because of global warming, but because of this emerging El Nino. So it is already kind of just increasing global temperatures over what they would have been in absence of this event. And it's nearly locked in a virtually guaranteed record temperature for 2027, even though we

haven't actually had that year unfold yet. It's no coincidence that the hottest years on record have almost always been El Nino years, right? When you take a warming trend and then you impose an anomaly on top of that, you're just going to get these outliers. And so these impacts are arguably already manifesting. So the South Asian monsoon was delayed.

that onset was delayed by this El Nino. And so while the kind of major teleconnexions and the peak strength associated with this event is anticipated to be in the winter, which is typically the case associated with El Ninos and just kind of the life cycle associated with them, its impacts are manifesting now and will continue to manifest in 2027. Our hometown is not a test tube. 90 miles northeast of Nashville, a battle for the future of America plays out in one small town. Developers with right-wing ties have purchased hundreds of acres of land. We need cities on a shining hill. This is Our Town, a podcast about what happens when a small town becomes the site of a social experiment and fights back. I guess you didn't move in on a bunch of dumb hillbillies now, did you? Listen to Our Town on the iHeart Radio app, Apple Podcasts, or wherever you get your podcasts.

Let's talk a little bit about the economic costs that we associate with an El Nino like I said up until your first answer

in this interview, I didn't know what an El Nino really was, but that certainly doesn't mean I've never said it or read it before. You know, it used to be on TV, you have a guest, he was like, oh, the upcoming El Nino. A joke stroking his beard, pontificating about the El Nino. Yes, mm-hmm. But it always seems like a sort of cyclical thing. It's like, okay, it hits every few years, and there's some costs, and then it goes back to normal. But... if there are whole books about the long-term consequences of any given El Niño, it sounds like at least in some serious ones there can be quite a tale, quite a, you know, ripple effects. When you go back and look at maybe some of the more recent big El Niños that we had, I think you mentioned there was one in the early 80s, how do you think about like tallying up? I don't know what the...

what the global economy, the price the global economy paid for it. Yeah. Such a good question. I want to call up a key collaborator. that I've had in this work, Chris Callahan, who's a professor at Indiana University and just a

stellar, stellar scholar. I couldn't have asked for a better PhD student to get to think about these hard things with. So, Chris had been thinking about El Nino and its kind of projection uncertainty and other contacts, meaning what does the future of El Nino look like in a world with global warming and how to understand that? Just be given that it's such an important part. of year-to-year variation in weather and climate globally. And it's essentially unknowable, right? It's very hard to know what El Nino is going to look like in 2050, what it's going to look like in 2100, and how that picture changes as a function of what your background warming happens to be. One of the things we had been investigating is just, well, we kind of had pivoted from thinking about future.

to saying, well, we live in a world where global warming has already occurred, we can simply document what has happened. And so much of what my lab group does here at Dartmouth is investigate retrospectively, kind of build that documentary evidence of here the cost of climate change to date. And in thinking about the cost of climate change to date, one of the things you have to do is isolate those costs. from the costs that would have happened anyway, such as the cost from El Nino, which is an innate form of climate variability, right? El Nino occurs in absence of people and their greenhouse gas emissions. And so isolating that cost and kind of partitioning it from the cost accounting of global warming to date became a pretty important exercise. So we're always in kind of detection and attribution science, trying to say, well, what is consistent with the world as it has been versus the world as it would have been absent global warming. And that means having a really clear picture of what we call internal climate variability, such as the variability from El Nino. And so we were interested in this cost accounting of El Nino just to clarify the cost of global warming to data. And when we went back and did this exercise using these causal inference techniques, so this is all observationally built, right? We can take data of variation.

in El Nino, which is essentially a random treatment, right? It is stochastically generated. While it has a life cycle against it, our global economies aren't necessarily shaping sea surface temperature anomaly patterns in places like the tropical Pacific, right? So it's effectively exogenous from our economy, right? And so that's really crucial in this

empirical setting of trying to identify, well, what's the effect of an El Nino? on an economy and how does that upward aggregate to a macroeconomic toll? What you can do is just take a bunch of data of something like gross domestic product, which is summarizing all economic productivity goods and services produced by an economy. We have pretty good data on that at the country scale for a pretty long amount of time. And you can look at differences in that economic growth. So how did economic growth vary from year to year?

So you're not looking at absolute levels of GDP or looking at changes in economic growth as embedded in changes in GDP. And then you can look at something like El Nino as a treatment. And then you have another source of variation, which is that not every country is equally linked to El Nino physically. El Nino is kind of remarkable in that it generates many different kinds of physical hazards from heat waves, to droughts, to floods, to landslides, to wildfires, right? And so in that sense, like its impacts, interestingly. look a lot like the wider impacts from global warming, which is a worthy discussion of itself, but we'll put that to the side for a second. And what you can do is you can exploit the kind of random variation we've observed in the real world of El Nino occurrence and El Nino magnitude, how that relates to the teleconnection, how tightly coupled locations, weather and climate is to El Nino as a phenomenon, and then see what accounts for the variation in economic productivity as measured in changes in GDP. And so that's what we did back in 2023 and published in a paper in Science, where we showed that actually we can detect very sizable deviations in economic growth.

from El Nino as a phenomenon and that scales in teleconnection strength, meaning countries that are most influenced climatically and from a meteorological standpoint by the phenomenon of El Nino are most impacted and negatively so. So Peru with road washouts, disrupting commodities from, you know, mine sites in the mountains to port induces a shock. And that shock doesn't seem to be what an economist might call a level effect, meaning a simple deviation that you then recover from in a subsequent year. What we find in this empirical research is that observationally, El Nino seems to systematically depress growth, meaning your country is growing at a rate and then an El Nino occurs and then you

are growing on a different trajectory. And so those losses just accrue ad infinitum in time. And that is a distinct finding than what was previously understood about El Nino. Right. So this is really important. I think most people are used to thinking about weather related shocks as the sort of short term thing that, as Justin just said, impacts the price level. But what the paper actually found was that there's this longer term impact on essentially output, like growth in the longer term.

Why is that like what exactly is happening here? Because again, you would expect like okay a road gets washed out. It's probably inconvenient It's probably expensive to replace but it does get you would think it does get replaced relatively quickly and a lot of these negative effects would subside relatively fast I think a one-off shock might look that way, right? A yield loss owing to a heat wave or drought in one part of the upper Midwest says nothing about major grain producing regions in Russia or cotton prices given growth in the Fergana Valley in Uzbekistan. These things are disparate enough in space that maybe they're not experiencing simultaneous shock. What I think is kind of interesting about El Nino and why El Nino is an interesting analog for the impacts of wider global warming, which is kind of a universally experienced, there's a high degree of correlation in the impacts of global warming. It's not simply a heat wave here. It's a heat wave here, here, here, and here, all at the same time, all experiencing loss at the same time, which is one of the reasons why I think it's this simultaneity.

of disparate regions experiencing hazards and experiencing losses that then can lead to these up aggregating market forces that systematically depress growth and alter where capital investment can go, the efficiency with which you're able to take labor and capital and mobilize it into productive parts of the economy. It is hard to do that when there are lots of hazards. being experienced around the world, right? So it's not one thing disrupting your supply chain. It's maybe a dozen things. And I think that's the distinction why, you know, the 9798 El Nino. was about 36 billion in economic losses as estimated at the time as a level fact, a shock and then a recovery. What we actually found is that there was empirical evidence for a systematic depression of economic growth. And over a five year period, that amounts to about 5.7 trillion in global economic losses instead by the year 2003, right, five years after the 97-98 event. What's really fascinating is we played

this game where we used what's called a perfect model framework where we simply ask the question if we know that the impact of El Nino is permanent and of this magnitude given the observational data we have could we prove that it's permanent? So these kinds of statistics rely on the ability to pull a signal from noise, which requires a certain amount of data. And that amount of data affects the claims you can make robustly. And so what we were trying to do there is say, even if we had a known effect, can we recover that known effect perfectly given our data record? And what we found is that we couldn't, which means that we cannot rule out the possibility that the economic impacts of El Nino at the global scale are permanent.

meaning that every time an El Nino occurs, we're just knocked off our growth trajectory that would have been absent that event, which is kind of remarkable, which implies that these costs kind of accrue ad infinitum in time. In our paper, we only went out to five and eight and 10 years of growth accounting, but it just says that the economic losses associated with these events are much higher than we understood before, which means it's just another kind of piece of evidence that shows that weather and climate matter for our economy and our climate system is endogenous to that economic growth. It is greenhouse gas emissions induced by economic growth that is altering our trajectory of climate. And our climate also has a finger on the scale of what constitutes our economy. And so that endogeneity I think is something really important to understand and represent.

particularly around policies related to climate variability and climate change. That was a fantastic answer. When you think about the impacts of this El Nino or the El Nino in the year 2050 or the El Nino in the year 2100, you know, so obviously like one, you know, presumably take a country like Peru, it's probably going to be richer in the year 2050, so it'll probably have more economic activity. It may also have better infrastructure in the year 2050, such that road washouts are less likely. Countries after, you know, I presume even since 82 or 97, there's been, you know, hardening against weather related shocks and so forth, as countries learn about these. How do you model that aspect, both the fact that, okay, there is more economic activity in the sort of temporally connected areas and also presumably

governments and companies get better about anticipating the El Ninos and strengthening their infrastructure such that they're not as vulnerable? It turns out it's really hard, right? The predictions of future are not falsifiable scientifically.

And so I hope Peru is wealthier in 2050 and 2100. Yeah, I mean, yeah, we didn't even talk about the fisheries implications. Yeah. One thing that's really, I think what you're kind of raising is this question of adaptation and resiliency. And the way I've been thinking about that is evolving, right? There's a story we can tell. that says, okay, if there are market inefficiencies or we're shown to be vulnerable societally in these particular ways, we're going to develop strategies to ensure resilience, right? Those Peruvian potato farmers, notice they were losing crops. You know, and they altered their sewing dates based on the visibility of the Pleiades overhead. Like that is an adaptation. And you can imagine autonomous adaptations, meaning individual level adaptations. You can also imagine society wide or planned adaptations, such as shoring up kind of resilience. And these things are linked to El Nino forecast, right? Like one of the things you're seeing.

You know you said at the outset Tracy of of all these financial planners thinking about the implications of El Nino for markets around the world and supply chains that is a form of kind of resilience strategy Building that that's occurring that isn't indelibly linked to the science of and so forecasting and why El Nino forecasts are so important, right? The 1982-83 El Nino, we didn't have the benefit of a tremendous amount of lead time and planning for its impacts, right? A tropical cyclone off in the ocean, right? Being able to track it and have sufficient time to get resources injected into the places you need and people out of harm's way, right? Like that requires lead time, which is why these types of forecasts are so essential.

because they allow us to pursue resilience strategies that we otherwise wouldn't have been able to. But there's a wider question of like, are we well adapted to climate variability we have? And shouldn't we expect that we'll be even better adapted to climate variability in the future? I would say the evidence is pretty mixed. There isn't a whole lot of evidence that we're well adapted to the climate we have. We are finding, documenting,

observationally, that the costs of El Nino are astounding, despite the fact that we've been exposed to it for hundreds and hundreds of years. This is not something that is novel. While the coming El Nino seems to be out of sample and the costs associated with it are really hard to anticipate because it is so out of sample. When I think about the cost of this forthcoming El Nino, I estimate somewhere around 10 trillion of global economic losses conservatively.

Wow. Over the next five years, right? Up to 14 trillion of economic losses if you take some of these tail. But I could only 70% of that, only, you know, about 10 trillion of that is in sample, meaning supported by existing data we have on El Nino as in economic responses. Our global economy is larger. And so these losses are associated not just with the magnitude of the El Nino, but the size of our global economy. There's just more stuff. to be impacted. What that tells me is that we are not optimally adapted to the climate we have. And so there isn't much evidence that we have that we are taking on board the feedback we're getting from the climate system about how things are going and pursuing strategies that really offset or mitigate losses in a coherent way. I think that's pretty telling just given the documentary exercise. That's not to say that we couldn't be.

better adapted. I just think that we are not well adapted and there isn't widespread evidence of adaptation to either El Nino as a form of climate variability or wider global warming since the impacts of global warming look a lot like the impacts of El Nino. And so I think there's like a bunch to kind of disentangle there, but when we're thinking about these future growth impacts on Enso, we need a counterfactual world of what economic growth trajectory would have been absent El Nino, but given global warming. And that is a picture of future that I think is really hard to envision. But there are economic tools that people use to create those counterfactual worlds. And then you can measure deviations as could be the case.

given what we know to be observationally true about the relationship between El Nino and an economy. I know you just said there aren't a lot of examples of successful adaptations for El Nino other than perhaps Peruvian potato farmers sowing their crops early. But give us some hope here or

some idea of what a successful adaptation could look like in an ideal scenario where this is an endogenous phenomenon as you just described where you have it basically happening all around the world simultaneously. What could we do to try to offset some of the economic impact? Gosh, that's, you know, it's such a great question. One of the ways I think about this is there's improving resiliency in global markets such that these weather and climate shocks don't alter.

prices and interest rates and all these things that are huge determinants of human well-being. Then there is also like the direct hazards themselves, right? A stronger than average atmospheric river event that triggers a landslide along Route 1 in California, creating a road washout that is a massive disruption to communities there. A wildfire, you know, that rages through the bush in Australia, killing trillions of animals and raising insurance premiums on kind of wildfire protection for homes there, right? There are all these kind of ways you can process trace how El Nino hurts an individual or a community. And so I think that protection of human well-being from approximate sense of like, are you okay given these hazards? That to me is a different question than how do you shore up international markets such that they are resilient to the impacts of El Nino. And I think just separating them is really, really valuable. I think the fact that people are aware of El Nino as a phenomenon and that it can induce these cascading shocks through agriculture or through mining or through the transport sector, and that all of those things all around the world can up aggregate to a macroeconomic toll, I think is important to know. And I don't know enough about how people think about these portfolios, but I know that risk diversification is a strategy to ensure returns. And I imagine that's something that people are embracing given modern portfolio theory or whatever. But I think this question of like, how do you protect people?

from hazard. That is a different question and requires different sets of warnings about the local manifestation of the hazards associated with El Nino. And to me, that means early warning systems. That means emergency management and response teams that are well placed and understand these consequences. That means that we have insurance markets that recognize the hazards, the unique geophysical hazards associated with something like El Nino and its regional dependencies and what it should imply for the

types of products those people need to protect the most capital intensive asset they have, which is their home. It's many different kinds of things. And so none of this is out of the realm of possibility. There's a lot of these things we do anyway, right? We build reservoirs to mitigate drought risk already. Never mind the fact that Lake Powell.

the central bank of the American West water economy is at a nearest record low. He knows his audience. Anyway, I guess the point is, is that, you know, I think there are lots of, there's lots of evidence that we, these adaptations are within the realm of possibility, but the constellation and configuration they take is kind of regionally dependent. And so I think kind of a coherent global strategy about managing the impacts of El Nino is an interesting test bed. for adaptation strategies that would lend itself to wider global warming. And that's something that I'm really intrigued by. So you talk about this idea that what you've discovered through your research is that El Ninos are not these sort of cycles that you bounce back from, and then you resume to trend growth, right? That they change their trajectory. But if El Ninos are something that are very consistent and predictable throughout history, such that there's evidence of farmers having tried to anticipate them going back over 500 years. Why should we not assume that it's seasonal? You could say, for example, that cold winter months depress economic growth, which they probably do. People don't go out as much and shop and travel as much.

in the middle of December and so we could say oh if there were no December's if it never got cold then in the counterfactual you know we would have had much higher growth which is true but also absurd because the premise of not having a winter is absurd if El Nino is something that will always and has always been with us why do we even imagine there being a counterfactual trend rate of growth in a world without El Niños? Yeah, I think that comes back to the kind of empirical identification strategy. We don't have the benefit of re-simulating the world as it has been and kind of the emergence of global society and the global economy absent El Niño. And so we rely on models to create those counterfactuals. We know that El Nino, while being cyclic in nature, it is not perfectly so. It is as good as a random treatment. And so we can separate the experience of the world into those associated with El Nino and those not. And we can

examine the differences in economic productivity under those two regimes. So that's essentially what we're doing empirically. It's akin to a medical control.

trial where you randomize a population into a treatment group that gets a drug and a control group that doesn't or receives a placebo and you compare medical outcomes. There are people that may look the same in the treatment and control group and there are people that may look different in the treatment and control group but if you randomized effectively those differences among individuals cancel out and the kind of efficacy of the drug itself is just the difference between those two groups. El Nino is as good as randomly assigned despite the fact that it is cyclic in nature. you can treat it as this natural experiment, this treatment, and compare it to the control of non El Nino years. And that's effectively what we're doing here when we're kind of talking about this counterfactual, right? We know what economic productivity looks like. And so you're right to say that, you know, economic productivity looks different in winter than in summer or in spring, right? And it looks different in the face.

of a tropical cyclone than it does in the face of, you know, a civil conflict or something else, right? Like these things have impacts, but what we're doing here is kind of exploiting natural stochastic variation in it, despite its cyclic nature. And what I think is remarkable, and what I'm kind of suggesting here is that despite the fact that El Nino is not a novel phenomenon, right? It is not something that we have never experienced before. which global warming is, right? Global warming is something humanity has never experienced and it is out of sample. And it is arguably making this El Nino that's unfolding, given the background warming, truly out of sample, truly out of sample event. Something humanity has never experienced before. And I'm saying, despite the fact that we should be inured to experiencing El Nino, we have limited evidence of widespread adaptation and resilience to it.

All right, Justin Manken, thank you so much for coming on Off-Logs. Yeah, thanks for having me. Joe, I found that conversation fascinating. Love talking about the weather and always. But I think the point about this idea that like, okay, El Nino is something that we have experience. It does come

with some regularity. And yet there seems to be some proof that it has this lasting impact, not just on the immediate economy, but on economic growth itself. is a very important one when you consider that, like, again, we're about to experience a potentially record-breaking non-nino and with the wider backdrop of climate change, right? Super interesting. I learned so much in that conversation that I didn't know this term that I just heard throughout my life that I really, oh, it gets warmer or something like that. I didn't really know what it was or why it was or what direction. I thought it was really interesting, okay, what makes and El Nino different from another random weather event, the global simultaneity of it is being key. Like, okay, other events you just bounce back from. I mean, everything must have some effect. You never like bounce back from anything perfectly back to trend. But the simultaneity, of course, we saw that during COVID and the ripple effects from that. And so when you have this weather pattern that is truly global in nature and sort of very, I guess, predictable in the sense that you know what regions they're all going to get hit, it sort of makes sense to think about it in these terms of like, oh, they're real long-term costs, especially if they're intensifying during a trend of global warming. Yeah, nowhere to run, nowhere to hide kind of idea. The other thing I was thinking, of course, back to some of our early agricultural episodes from this year, all of this is happening against a backdrop of higher fertilizer prices and lower supply, like you still have, I think.

for 30%, 40%. I can't remember the exact numbers of global fertilizer supply stuck in the Strait of Hormuz. And that can't be helpful either when you're talking about... the biggest El Nino on record. Yeah. No, it's going to be a, well, now we are well armed. Maybe we're not well adapted for the El Nino, but at least we are at least well armed to understand news stories that will be emerging over the weeks. And the cell-side analysts. And the cell-side analysts notes in the months ahead. All right. Shall we leave it there? Let's leave it there. This has been another episode of the Odd Thoughts podcast. I'm Tracy Allaway. You can follow me at Tracy Allaway.

And I'm Joe Wiesenthal, you can follow me at the stalwart. Follow our producers, Carmen Rodriguez at Carmen Arman, Dash Obenet at Dash Bot, Kel Brooks at Kel Brooks, and Kevin Lozano at Kevin Lloyd Lozano. And for more OddLots content, go to Bloomberg.com slash OddLots for the

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